

(MATTHEW) YIZHENG HE

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Education

McGill University	2023 – 2027 (expected)
B.A. & Sc. Honours Cognitive Science (Computer Science stream), Minor Mathematics	CGPA: 4.0/4.0 (Top 5%)

Research Experience

Masset Lab <i>Research Intern</i>	2024–Present
↪ Prof. Paul Masset	McGill University, Mila–Quebec AI Institute

Projects involved:

- **Simultaneous detection and estimation in olfactory sensing (Sep. 2024–Present)**
(Recurrent neural networks; Constrained optimization; Bayesian inference; Mirror-Langevin descent)
Designed, coded, and ran scaling experiments on HPC clusters; refined model variants to improve performance.
First author on the accepted abstract and co-first author on the associated preprint (under review at PNAS).
- **Population structure of reward-induced remapping in hippocampal CA1 (Apr. 2025–Present)**
(Representation similarity analysis (RSA); Tensor decomposition; Dimensionality reduction techniques)
Built data pipelines for preprocessed data; performed exploratory data analysis and developed metrics to quantify neural population structure and reward-related remapping. Made figures for the submitted COSYNE abstract.
- **Piecewise deterministic spiking samplers (Nov. 2025–Present)**
(Piecewise-deterministic Markov processes; MCMC sampling algorithms; Spiking neural networks)
Participating in both simulation-based and theoretical investigation.

Research Interests

Computational & Theoretical Neuroscience; Machine Learning Theory

Preprint

Chen Jiang*, **Matthew Y. He***, Venkatesh N. Murthy, Cengiz Pehlevan, Jacob A. Zavatone-Veth[†], and Paul Masset[†] (2025). “Simultaneous detection and estimation in olfactory sensing”. In: *bioRxiv*. (Under review at PNAS). doi: [10.1101/2025.11.01.686013](https://doi.org/10.1101/2025.11.01.686013).

* Equal contribution. [†] Co-supervision.

Conference Abstract

Matthew He, Chen Jiang, Venkatesh Murthy, Cengiz Pehlevan, Jacob Zavatone-Veth, and Paul Masset (2025). “Simultaneous detection and mapping in the olfactory bulb”. In: *Computational and Systems Neuroscience (COSYNE)*.

Chen Jiang, Adel Halawa, **Matthew He**, and Paul Masset (2026). “Population structure of reward-induced remapping in the hippocampal CA1”. In: *Computational and Systems Neuroscience (COSYNE)*.

Awards & Distinctions

Faculty of Science Scholarship	2025-2026
Established in 1992 by the University to provide awards based on academic achievement to students in the top 5% of the Faculty of Science.	

Dean’s Honour List	2023-2025
Top 10% of students in the Faculty of Science based on the combined GPA for the fall and winter terms.	

Science Undergraduate Research Awards	2025
\$8950 CAD for 15 weeks of full-time research in the summer of 2025.	

SUS Ambassador Fund	2025
\$750 CAD for traveling to present at COSYNE conference 2025.	

Tomlinson Undergraduate Award	2025
Mentoring students in a molecular and cellular biology course.	

Conference & Workshop Attendance

Mathematical Foundation for Data Science	<i>Stochastic Optimization; High-dimensional Statistics</i>	May–June 2025
↪ Centre de Recherches Mathématiques, Montreal, Canada		
Computational and Systems Neuroscience (COSYNE) 2025	<i>Conference & Workshop</i>	April 2025
SynBio Collective Wet Lab Workshop Series	<i>Theme: Expressing recombinant genes in bacteria</i>	January–April 2024
↪ McGill iGEM Team		

Selected Coursework

Graduate-level	Math & CS (selected)	Expected by Summer 2026
Stochastic Processes*	Nonlinear Optimization	Convex Optimization*
Computational Neuroscience*	Hon. Applied Linear Algebra	Theory of Machine Learning*
Numerical Analysis*	Hon. Nonlinear Dynamics	Representation Learning*
	Hon. Algorithms & Data Structures	
	Probability & Statistics (2 terms)	
* Graduate level.		

Skills

Scientific Computing	Python (PyTorch, JAX), MATLAB, Linux, HPC clusters (SLURM)
Mathematical Areas (beyond coursework)	Bayesian inference, sampling methods, tensor decomposition, physics-informed neural networks (PINNs)
Neuroscience Knowledge	Hippocampus, olfactory bulb, midbrain dopaminergic neurons; general systems neuroscience knowledge learned through courses